

NEW YORK STEWART, NY, USA

WMO: 725038

Lat: 41.500N

Lon: 74.100W

Elev: 491

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.7	(c) 7.8	(d) -8.5	(e) 3.6	(f) 6.9	(g) -4.4	(h) 4.4	(i) 11.7	(j) 28.6	(k) 28.3	(l) 26.3	(m) 22.9	(n) 7.3	(o) 290	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	90.3	72.1	87.6	71.5	84.1	69.6	75.7	84.7	74.1	82.6	72.7	80.6	10.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	124.9	79.4	72.1	120.7	78.5	70.2	113.3	77.0	39.4	84.8	38.0	82.5	36.5	80.8	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.6	(b) 20.8	(c) 18.9	(e) -4.5	(f) 94.8	(g) 6.7	(h) 4.0	(i) -9.4	(j) 97.7	(k) -13.3	(l) 100.1	(m) -17.0	(n) 102.4	(o) -21.9	(p) 105.3		
			DB													
			WB													
				-3.4	78.3	5.3	2.6	-7.2	80.2	-10.2	81.7	-13.2	83.2	-17.0	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	25.7	29.9	37.8	49.0	58.4	68.6	73.0	71.6	63.5	51.9	41.7	32.2		
	DBStd	18.00	11.33	9.53	9.03	8.35	7.02	6.54	5.28	5.46	6.46	7.31	8.34	8.70		
	HDD50	2726	755	564	391	115	8	0	0	0	0	67	271	555		
	HDD65	6038	1218	983	844	486	228	37	4	10	103	410	698	1017		
	CDD50	2868	2	1	14	84	270	558	712	670	406	124	23	4		
	CDD65	704	0	0	1	5	25	145	251	215	59	3	0	0		
	CDH74	5960	0	0	12	103	320	1212	2134	1692	441	44	2	0		
	CDH80	1873	0	0	3	30	92	376	744	535	90	3	0	0		
(14) Wind	WSAvg	8.4	9.4	9.0	9.8	9.7	8.8	7.6	7.3	6.6	7.2	7.9	9.0	9.1		
(15) Precipitation	PrecAvg	43.7	3.0	2.4	3.4	3.7	4.2	4.3	3.9	3.8	4.0	3.5	3.7	3.2		
	PrecMax	58.9	9.8	4.6	7.6	8.8	10.2	11.5	13.6	9.6	9.9	17.9	7.6	7.5		
	PrecMin	28.3	0.2	0.2	0.4	1.2	1.2	0.4	0.8	0.6	0.7	0.0	0.8	0.8		
	PrecStd	7.6	1.8	1.0	1.5	1.5	2.1	2.4	2.3	2.0	2.4	2.8	1.7	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	60.7	74.9	84.0	88.4	90.9	95.5	93.3	87.6	79.0	68.5	63.5		
		MCWB	55.9	51.9	58.8	64.9	66.7	73.2	74.6	72.9	70.2	62.4	60.4	54.0		
	2%	DB	52.4	52.4	64.0	75.3	82.0	88.0	90.7	89.0	82.0	73.3	64.2	54.6		
		MCWB	49.2	42.3	51.9	58.3	65.1	72.4	73.0	71.6	67.2	61.0	57.5	49.2		
	5%	DB	45.7	47.6	58.0	68.4	76.9	84.2	87.6	85.7	79.2	69.6	60.7	49.4		
		MCWB	42.2	41.7	47.9	55.7	62.0	70.4	72.2	70.3	65.6	59.8	53.3	45.1		
	10%	DB	41.1	43.5	52.2	63.6	72.2	81.3	83.9	82.2	75.3	64.3	55.2	45.1		
		MCWB	37.3	37.2	44.4	51.4	58.9	68.7	70.5	68.7	64.8	56.2	48.8	40.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	53.2	59.9	66.8	70.4	77.0	79.6	76.8	74.4	68.0	63.0	54.6		
		MCDB	59.9	58.6	73.1	81.9	81.7	85.5	91.0	85.4	79.8	72.8	65.3	60.2		
	2%	WB	49.9	46.2	54.7	60.6	67.4	74.7	76.3	74.8	71.3	63.7	59.7	49.7		
		MCDB	52.0	49.6	61.8	71.2	77.2	83.6	85.4	83.7	77.3	69.7	64.3	53.9		
	5%	WB	43.5	41.3	49.9	57.8	64.6	72.8	74.3	73.4	69.3	60.6	54.7	45.8		
		MCDB	45.9	46.7	57.6	66.6	73.4	81.0	82.8	81.8	74.0	67.0	58.5	48.7		
	10%	WB	37.5	37.7	45.2	53.9	61.9	70.8	72.7	71.9	67.2	57.8	50.5	40.3		
		MCDB	40.8	42.8	51.2	60.7	69.8	78.2	80.9	79.2	73.3	63.6	55.1	44.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	17.2	17.9	20.6	20.6	19.6	20.0	19.6	20.1	19.6	16.9	14.9		
		MCDBR	19.2	23.3	26.1	29.1	28.6	23.0	23.8	23.3	23.5	27.1	22.1	20.9		
		MCWBR	16.4	17.4	17.1	16.4	14.4	11.3	9.6	9.5	11.5	15.4	16.2	17.0		
	5% WB	MCDBR	18.8	19.9	25.0	25.0	23.7	20.4	19.6	18.2	18.7	22.2	18.7	18.9		
		MCWBR	17.4	16.7	18.5	16.3	13.7	11.4	9.9	8.9	11.7	15.5	16.4	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.318	0.344	0.363	0.405	0.464	0.473	0.453	0.381	0.352	0.318	0.301		
	taud		2.418	2.381	2.394	2.392	2.296	2.166	2.154	2.200	2.377	2.470	2.518	2.500		
	Ebn at Noon		270	282	287	288	277	259	256	257	271	267	263	262		
	Edh at Noon		25	30	34	36	41	47	47	43	34	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		564	841	1152	1466	1637	1780	1849	1619	1340	899	625	461		
	RadStd		42	81	89	156	163	167	103	105	140	97	59	48		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+108

Nomenclature: See separate page